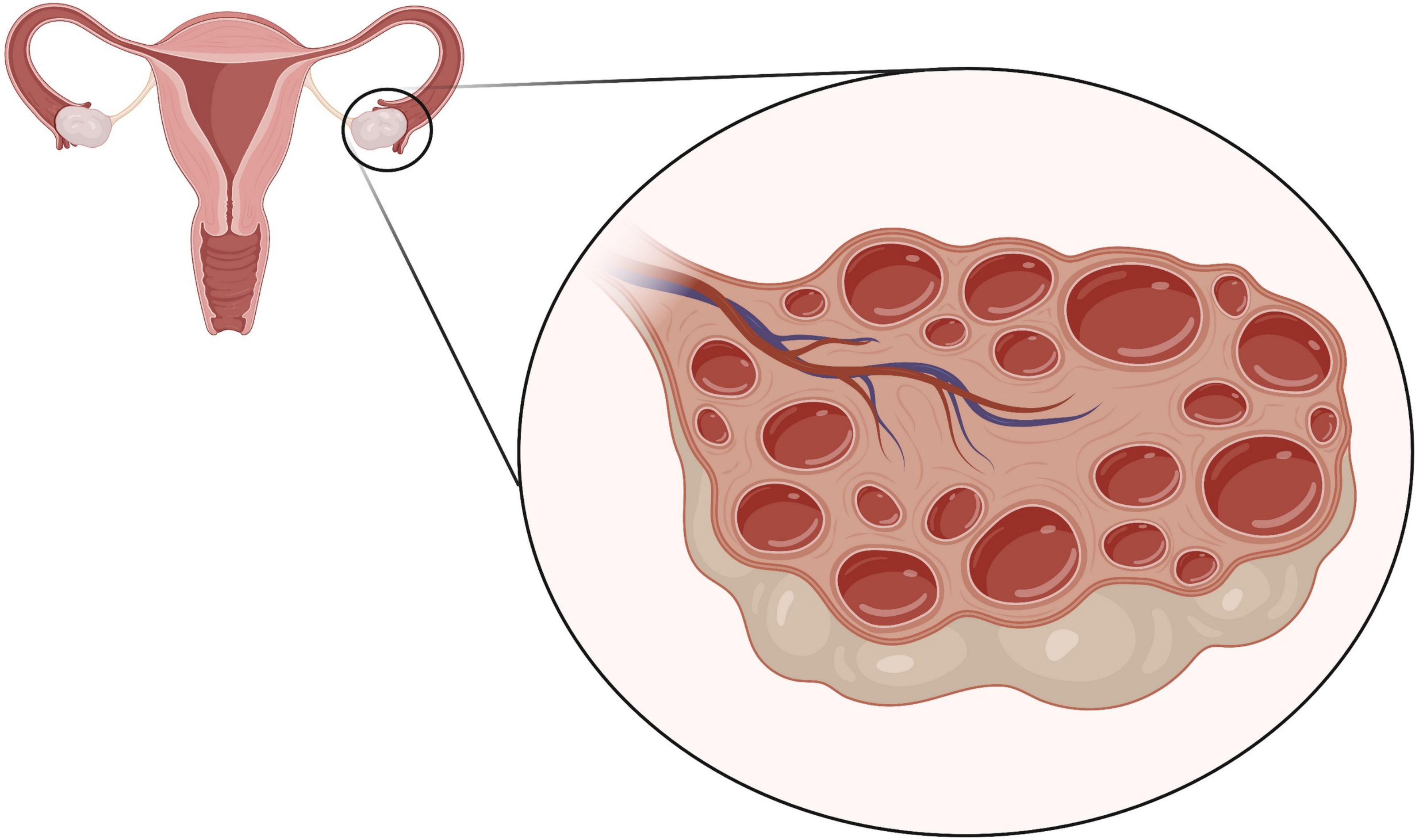
Diagnosis

DDx and Eval

Associated Co-Morbidities

Cases

Polycystic Ovarian Syndrome (PCOS)



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How do we diagnose PCOS?

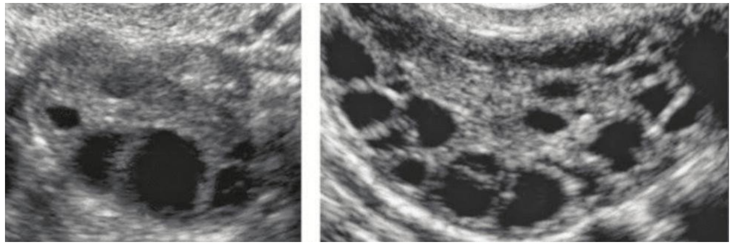
We use the **Rotterdam Criteria** to diagnose PCOS.

2 of the 3 criteria are needed for diagnosis

1 Hyperandrogenism → Clinical OR Biochemical
- Clinical: acne, hair loss, hirsutism
- Biochemical: ↑ circulating androgen levels

2 Menstrual irregularities → Chronic anovulation
- Oligomenorrhea (cycle >35d)
- Amenorrhea (no menses x 3 mos)

3 Polycystic Ovaries → >12 follicles and/or volume > 10 cc (seen on pelvic ultrasound)



Normal ovary   Polycystic ovary

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PCOS is a diagnosis of exclusion!

Many conditions mimic signs and symptoms of PCOS.
What other conditions are on your differential?

1 Hyperandrogenism

- **Congenital adrenal hyperplasia**
- Acromegaly
- Androgen secreting tumors
- Cushing's syndrome

Total testosterone
+ 17-OH progesterone

Evaluation?

Midnight salivary cortisol

2 Menstrual irregularities

- **Thyroid disorders**
- **Hyperprolactinemia**
- **Endometrial causes** (polyp, adenomyosis, leiomyoma, malignancy/hyperplasia)
- Perimenopause/ ovarian failure
- Pregnancy
- Hypothalamic hypogonadism

TSH, fT4
Prolactin

Evaluation?

± LH/FSH

β-HCG

3 Polycystic Ovaries

- **Normal cycle**
- Exogenous testosterone use

What comorbidities are associated with PCOS?

Mood disorders

↑ risk of anxiety, depression, ED

Screen Management? d GAD

Hyperandrogenism (1º feature)

↑ androgens → ↑ facial hair, acne

OCPs ± sp Management? ndrogens.
Cosmetic emoval)

Cardiovascular risk

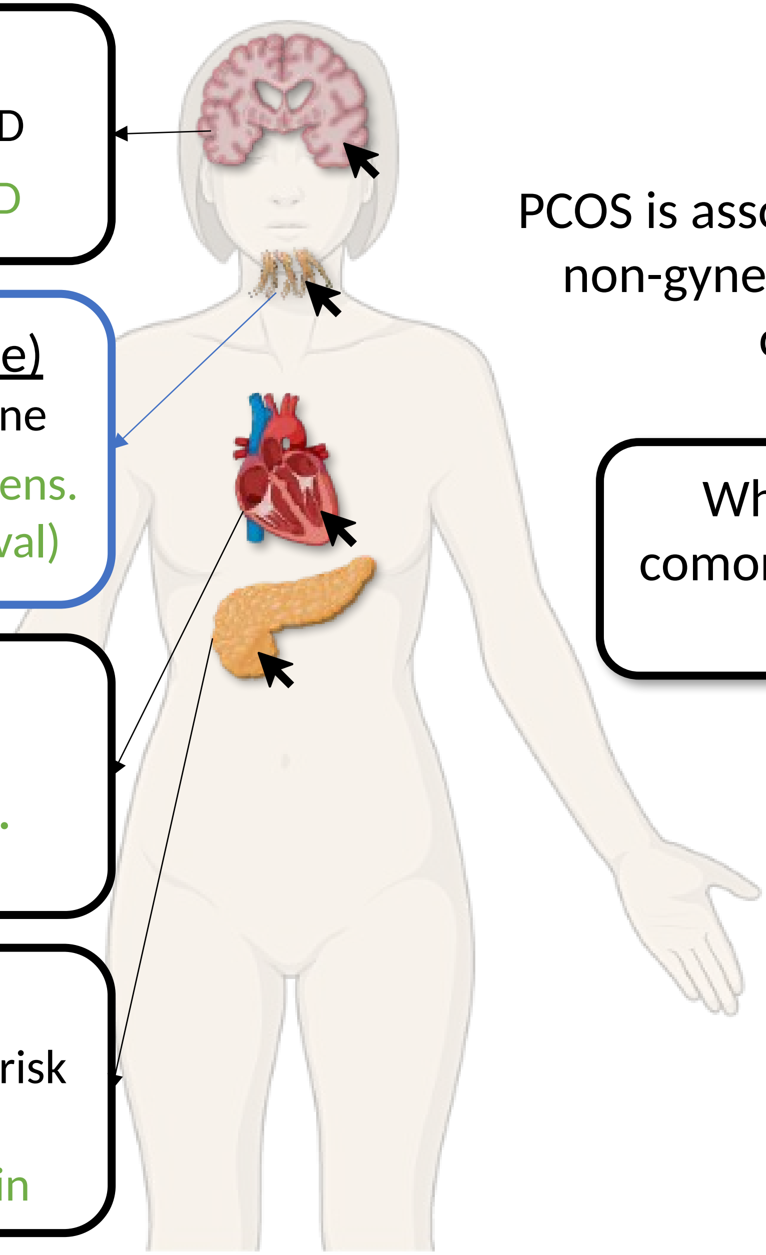
↑ lipids, ↑ CV events

Screen Management? **panel.**
Treat CV risk factors.

Insulin resistance

Common regardless of weight, ↑ DM risk

Scree Management? **g BG.**
Lifestyle, I etformin



PCOS is associated with wide array of non-gynecologic and gynecologic comorbidities.

What **non-gynecologic** comorbidities are associated with PCOS?

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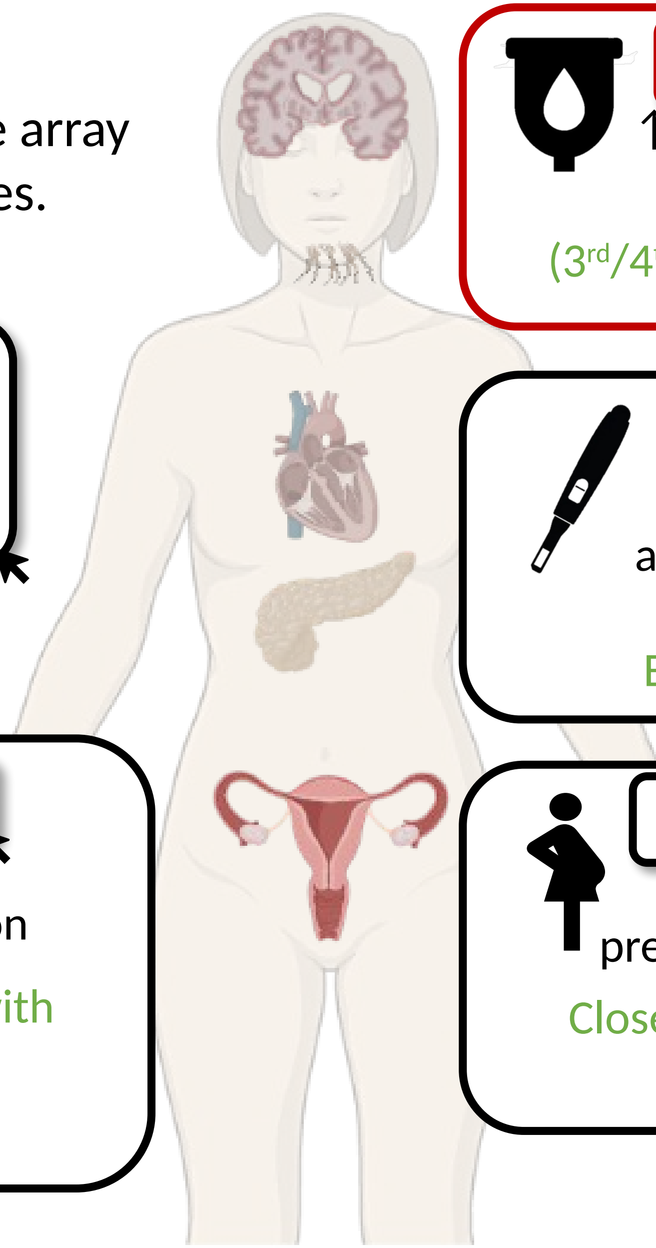
- Diagnosis
- DDx and Eval
- **Associated Co-Morbidities**
  - Non-Gyn
  - **Gyn**
- Cases

What comorbidities are associated with PCOS?

PCOS is associated with a wide array of gynecologic comorbidities.

What **gynecologic** comorbidities are associated with PCOS?



Menstrual irregularities

↑ irregular/disruptive bleeding

OC Management? es

(3rd/4th G progestins are less androgenic)

Infertility

Most common cause of anovulatory infertility in the US

D Management? ls

Early r Management? or REI

Endometrial hyperplasia

3x ↑ risk of endometrial cancer due to chronic anovulation

Promote endometrial turnover with OCPs, Management? cyclic progesterone

Pregnancy complications

Early pregnancy loss, GDM, pre-eclampsia, and pre-term labor

Close mon Management? f condition develops

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Case 1a

Olga is a 24 yo woman who presents with irregular periods since menarche. Her periods occur every 45 to 60 days.

Family history is notable for type II DM.

What additional history or exam findings can help build your differential?

e has **oligomenorrhea** (a form of abnormal uteri bleeding). Causes and associated signs/symptoms include:

Hirsutism	PCOS, hyperandrogenism
Breast discharge	Prolactinoma
Hot flashes	Premature ovarian failure
Weight gain	Cushing's, hypothyroidism
Weight loss	Stress, malnutrition, hyperthyroidism
Sexually active	Pregnancy, STIs
Abnormal pelvic exam	Structural cause (e.g., polyp, fibroid)

…She has acne and some hair growth on her chin. She reports slow weight gain. Otherwise, she is not sexually active.

What are your next steps in evaluation?

- **Beta-HCG** (rule out pregnancy in all reproductive age women)
- **Transvaginal US** (rule out structural causes)
- **TSH, prolactin, ~~FSH~~**
- **Total testosterone** (signs of hyperandrogenism, ± 17-OH progesterone, ± SHBG, ±DHEAS)

Case 1b

Olga returns for a follow-up visit to review her labs and imaging results.

Labs: TSH and prolactin are normal. Total testosterone is elevated.

Transvaginal ultrasound 15-20 small follicles in both ovaries. Otherwise, normal.

What is her diagnosis?

PCOS. She meets all 3 Rotterdam criteria: oligomenorrhea, biochemical hyperandrogenism, and polycystic ovaries. Other causes of oligomenorrhea ruled out. Remember, PCOS is a diagnosis of exclusion!

What additional non-gynecologic comorbidities should you screen for?

- Psych → MDD, GAD screen
- Insulin resistance → HbA1c, fasting BG
- CV risk → fasting lipids

What gynecologic comorbidities is she at risk for?

- Infertility
- Pregnancy complications
- Endometrial hyperplasia and higher risk of endometrial cancer

What options does she have for endometrial protection?

Combined oral contraceptives, progestin-containing IUDs, and cyclic progesterone all can be used to decrease risk of endometrial hyperplasia.

2023 Guideline Update — Rotterdam framework unchanged

• The 2023 International Evidence-Based Guideline for PCOS (Teede et al.) updates the 2018 version the deck builds on. Rotterdam (2 of 3) and "diagnosis of exclusion" are unchanged.
• NEW diagnostic option: anti-Müllerian hormone (AMH) may substitute for pelvic ultrasound to establish polycystic ovarian morphology in ADULTS (not adolescents). For irregular cycles + hyperandrogenism (~70% of cases), neither US nor AMH is needed — those two criteria already make the diagnosis.
• Adolescents: require BOTH hyperandrogenism AND ovulatory dysfunction; ultrasound and AMH are NOT recommended (<8 yr post-menarche) due to poor specificity.
• Strengthened recognition of the metabolic/CV burden, obstructive sleep apnea, and the very high prevalence of psychological features (screen for depression, anxiety, and disordered/binge eating).
• Therapy: lifestyle first; metformin for metabolic/weight/cycle benefit; combined OCPs ± anti-androgen for hyperandrogenism/cycle control. Anti-obesity pharmacotherapy incl. GLP-1 receptor agonists may be CONSIDERED for weight in adults (evidence still limited — high-priority research area).

– Sources: Teede HJ et al., 2023 International Evidence-Based Guideline for PCOS (Hum Reprod / JCEM / Fertil Steril 2023); Goldberg, Obes Rev 2024 (anti-obesity agents).

Take-Home Points

• PCOS is defined by the Rotterdam criteria — hyperandrogenism (clinical or biochemical), menstrual irregularity/ovulatory dysfunction, and polycystic ovaries on ultrasound (or, in adults, AMH per 2023 guideline). Two of three are required.
• It is a DIAGNOSIS OF EXCLUSION — rule out other causes first: total testosterone (± 17-OHP, SHBG, DHEAS), TSH/free T4, prolactin, β-hCG; consider midnight salivary cortisol or LH/FSH per clinical picture.
• Screen for the comorbidities, not just the diagnosis: insulin resistance/T2DM (A1c, fasting glucose or 2-h OGTT, q1–2y), cardiovascular risk (BP, fasting lipids), mood/eating disorders (PHQ-9, GAD-7), OSA, and NAFLD.
• Protect the endometrium: chronic anovulation → 3× endometrial cancer risk. Ensure a withdrawal bleed at least every 90 days (OCPs, progestin IUD, or cyclic progesterone). Routine US screening for endometrial cancer is not recommended.
• PCOS is the most common cause of anovulatory infertility — discuss fertility goals early; 5–10% weight loss improves ovulation; refer to OB/Gyn or REI; clomiphene/letrozole for ovulation induction.

Attribution & Changelog

- Original: "Polycystic Ovary Syndrome," TeachIM.org, published May 2022 — Nikki Zarling MD, Natalie deQuillfeldt MD, Molly Brett MD; Section Ed. Adelaide McClintock MD; Exec. Ed. Yilin Zhang MD.
  - https://teachim.org/teaching_material/pcos/ — CC BY-NC 4.0 (attribution required, non-commercial). Optimized June 2026. Deck: Teach-IM-PCOS-yz-6.2-1.pptx (page embed URL).
- **Correction to this copy (all click-to-reveal animations & BioRender graphics preserved):**
  - Slide 3 (menstrual-irregularity evaluation): "± LH/FH" → "± LH/FSH" (the page text and standard workup are LH and FSH).
- Appended: this changelog + 2023 Guideline Update + Take-Home slides. No original content removed; 7 original slides (1 hidden learner worksheet) → 10.
- Companions copied & linked (reviewed, unchanged): PCOS-Teach-IM-Guide-yz-6.2-1.docx (facilitator guide), PCOS-Learner-Handout.pdf. Package: HTML, OnePager PDF, Anki TSV.
- Educational use only — not medical advice; clinician-review before teaching.